

KPP Validation Report

MITgcm vs Python Port Comparison

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MITgcm Reference:

File: mitgcm_kpp_outputs_lab_sea_6mo.nc
UUID: 58aa5873-19ba-40d6-960b-b54d9fe50390

Python Port Output:

File: python_kpp_outputs_lab_sea_6mo.nc
UUID: 58aa5873-19ba-40d6-960b-b54d9fe50390

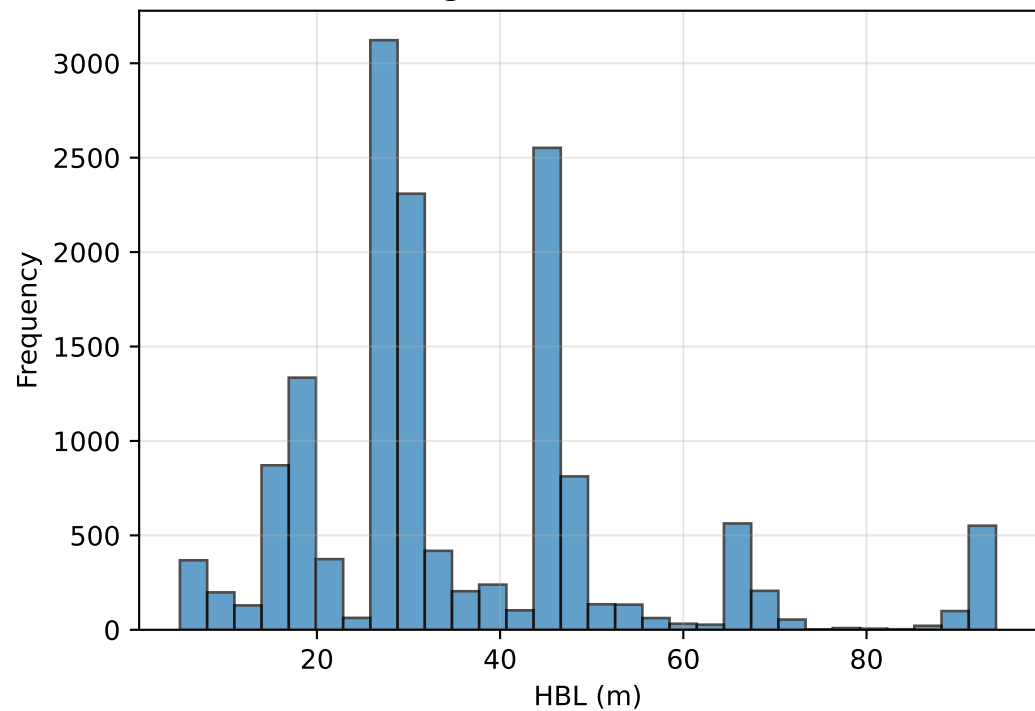
Provenance: ☒ UUIDs match

Timesteps: 100
Grid: 20 × 16 × 23

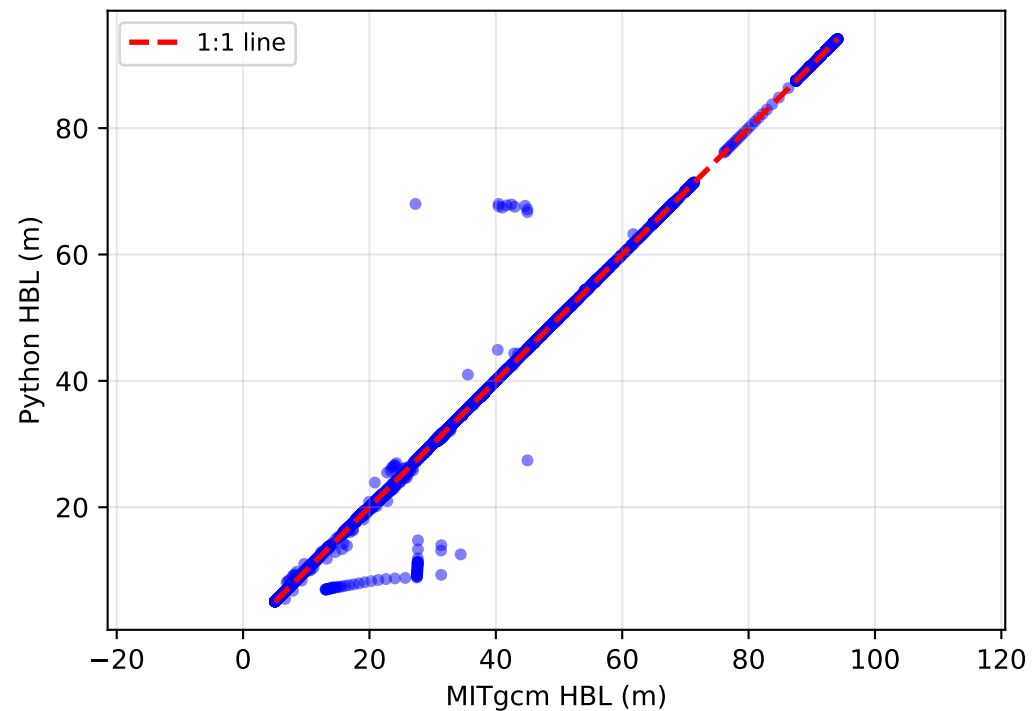
Note: Python output covers only the first 100 of 4368 timesteps; MITgcm restricted to the same leading subsample for this comparison.

Boundary Layer Depth (HBL) Comparison

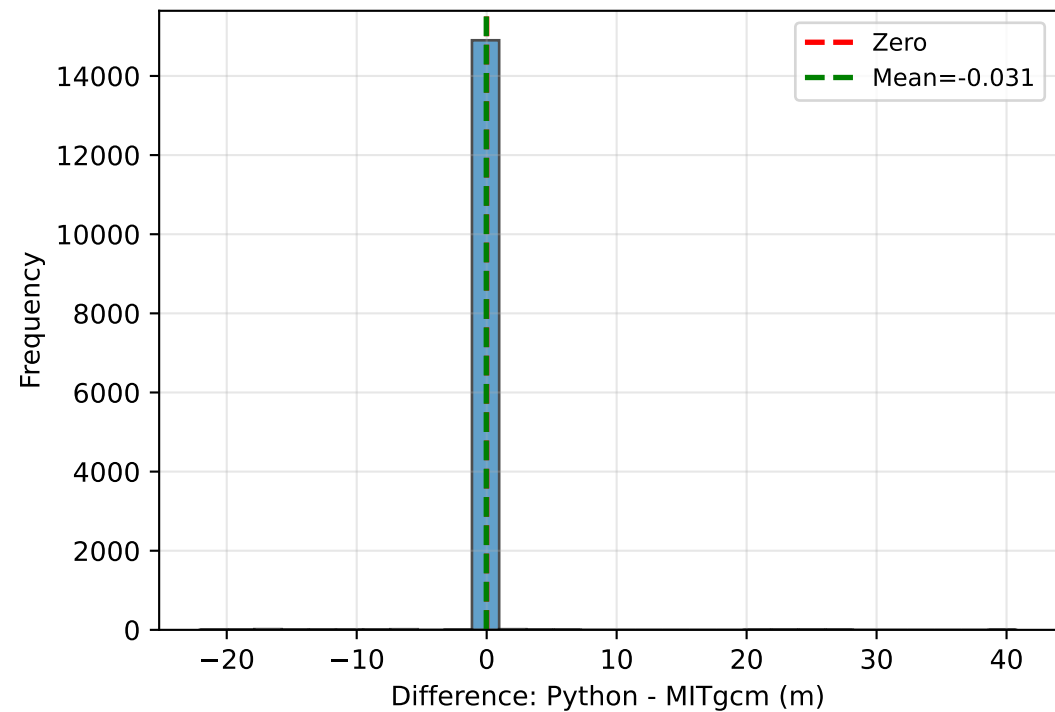
MITgcm HBL Distribution



HBL Scatter Plot



HBL Difference Distribution



HBL Statistics:

Mean |diff|: 0.097158 m
Max |diff|: 40.723362 m
RMS diff: 1.136860 m

Mean rel error: 0.4092%
Median rel error: 0.0082%

Points compared: 15000

Range:
MITgcm: 5.1 – 94.1 m
Python: 5.1 – 94.1 m

Assessment:

△ □ CHECK

Mixing Coefficients Statistics

Active Mixing Regions (coefficient > 1e-6 m²/s)

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VISC_AZ: □ EXCELLENT

Median relative error:	0.000000%
Mean relative error:	19.547332%
Max relative error:	51230.753624%
95th percentile error:	20.322954%
Median absolute diff:	0.000000e+00 m ² /s
Points compared:	171000

DIFF_KZ_S: □ EXCELLENT

Median relative error:	0.000000%
Mean relative error:	26.316335%
Max relative error:	131053.604266%
95th percentile error:	13.976801%
Median absolute diff:	0.000000e+00 m ² /s
Points compared:	171000

DIFF_KZ_T: □ EXCELLENT

Median relative error:	0.000000%
Mean relative error:	26.316335%
Max relative error:	131053.604266%
95th percentile error:	13.976801%
Median absolute diff:	0.000000e+00 m ² /s
Points compared:	171000

GHAT: □ EXCELLENT

Median relative error:	0.091025%
Mean relative error:	0.415548%
Max relative error:	286.375407%
Points compared:	34540

Validation Summary & Assessment

OVERALL VALIDATION RESULT:

Status: ☒ NEEDS REVIEW

COMPONENT RESULTS:

HBL (Boundary Layer Depth):	☒ CHECK
Max difference:	40.723362 m
Mean relative error:	0.4092%
VISC_AZ:	☐ EXCELLENT
Median relative error:	0.000000%
DIFF_KZ_S:	☐ EXCELLENT
Median relative error:	0.000000%
DIFF_KZ_T:	☐ EXCELLENT
Median relative error:	0.000000%
GHAT:	☐ EXCELLENT
Median relative error:	0.091025%

PROVENANCE:

UUID Match:	☐ Yes
MITgcm UUID:	58aa5873-19ba-40d6-960b-b54d9fe5...
Python UUID:	58aa5873-19ba-40d6-960b-b54d9fe5...

ACCEPTANCE CRITERIA:

☐ EXCELLENT:	HBL < 0.1m, mixing < 0.1% error
☐ GOOD:	HBL < 1.0m, mixing < 1.0% error
☒ CHECK:	Exceeds thresholds - investigate